
BIRD MONITORING PROJECT

Explanatory Video Speaking Script

A natural, professional talk-track for presenting the data-cleaning workflow, Streamlit dashboard, charts, insights, and limitations.

Project scope	Data manipulation, cleaning, feature engineering, dashboard analysis
Dataset scale	33,251 observations across 78 unique species
Core tools referenced	Python / pandas / interactive dashboard analysis
Speaking formula	Question -> What I did -> What I found -> Why it matters

Use this script as a speaking guide rather than reading every sentence exactly. The language is written to sound natural while presenting each slide.

Source basis: the supplied Bird Monitoring Project report and dashboard analysis. Key figures and interpretations are preserved from the supplied material.

Slide 1 - Project Introduction

Hello everyone, my name is Fateh Mohammed Sayyed, and today I am going to explain my Bird Monitoring Project. The main objective of this project was to take bird-monitoring data, clean and prepare it properly, and then use that prepared data to build an interactive dashboard for analysis.

The project is divided into two major parts: first, data manipulation and cleaning, and second, dashboard analysis and insights. The overall flow is source data, cleaning and validation, feature engineering, dashboard, and finally insights.

Natural interview-style questions

Question: Why did I separate data cleaning from visualization?

Answer: Visualization is only useful when the underlying data is reliable, so I first made sure that the dataset was properly prepared before creating analysis or charts.

Slide 2 - Project Data-Manipulation Workflow

Now let us look at the data-manipulation workflow. Before creating any visualization, I first created a clean analytical dataset. I loaded the data, inspected the schema and data types, profiled missing values, standardized the columns, removed empty rows and columns, converted dates, cleaned text and blank values, removed duplicates, created time features, and finally validated the output. The main idea is that I do not directly jump from raw data to charts. I first make the data consistent and reliable, and only then use it for analysis.

Natural interview-style questions

Question: What is the main idea behind this workflow?

Answer: The main idea is to treat data cleaning as a controlled transformation and make the final dataset safe and reproducible for downstream analysis.

Speaking tip: explain the workflow as a sequence. Do not rush through the individual cleaning steps.

Slide 3 - Loading and Initial Inspection

The first practical step was loading the dataset using pandas. I used `read_csv` to load the prepared CSV file into a DataFrame. After loading the data, I checked the shape of the dataset to understand the number of rows and columns, and I also displayed the first few records. This initial inspection is important because before modifying the data, I want to know what the incoming dataset actually looks like. At this stage, I am not performing analysis yet; I am only trying to understand the structure of the data.

Natural interview-style questions

Question: Why do we inspect the data before cleaning it?

Answer: Because we should understand the original structure before making transformations. Otherwise, we may make changes without knowing what problem we are actually solving.

Slide 4 - Schema and Data-Type Inspection

After loading the dataset, the next step was checking the schema and data types. I checked the column names, the data type of each column, and the overall DataFrame information. This is especially important for fields such as dates. If a date column is stored as text instead of a proper datetime format, I cannot reliably extract year, month, or weekday from it. So this step helped me identify which columns were already suitable for analysis and which ones needed conversion.

Natural interview-style questions

Question: Why are data types important in data analysis?

Answer: Because different analytical operations require the correct data type. For example, mathematical operations require numeric fields, while time-based analysis requires proper datetime values.

Slide 5 - Missing-Value Profiling

The next step was missing-value profiling. I calculated the number of missing values for each column and also calculated the missing percentage. I sorted the results so I could quickly identify which fields had the highest amount of missing data. The important point here is that I did not blindly replace or delete missing values. First, I measured the missingness and understood where it occurred. Only after understanding the pattern could I decide how the data should be handled.

Natural interview-style questions

Question: Why should we not simply replace every missing value?

Answer: Because missing values do not always mean the same thing. Automatically filling everything can introduce incorrect information. We first need to understand the reason for the missingness and its effect on analysis.

Slide 6 - Core Data Cleaning

Now we come to the main data-cleaning stage. First, I created a copy of the original DataFrame so that the original dataset would remain unchanged. Then I standardized the column names to make them consistent and easier to work with. I removed completely empty rows and columns because they do not provide analytical value. After that, I converted the date field into a proper datetime format. I also cleaned text fields by removing unnecessary spaces and converted blank strings into proper missing values. Next, I checked for duplicate rows and removed duplicate records. Finally, because temporal analysis depends on valid dates, records without usable dates were removed for that analysis.

Natural interview-style questions

Question: Why did I remove duplicate records?

Answer: Because duplicate observations can artificially increase counts. If the same observation appears more than once, the dashboard may show an incorrect number of bird observations.

Question: Why did I standardize column names?

Answer: To make the dataset easier and safer to work with. Consistent column names reduce errors when writing filtering, grouping, and transformation logic.

Speaking tip: emphasize that each cleaning operation solves a specific data-quality issue.

Slide 7 - Post-Cleaning Validation

After cleaning the data, I did not immediately move to the dashboard. I performed another validation step. I recalculated the missing values, checked the missing percentages again, verified that duplicate rows remaining were zero, and displayed the cleaned records. This is an important part of the project because cleaning is not complete just because we have written the cleaning code. We need to verify that the transformation actually produced the expected result.

Natural interview-style questions

Question: How do I know my cleaning process worked?

Answer: I compare the data quality before and after cleaning. I check remaining missing values, duplicate records, and sample records again. This gives me confidence that the final dataset is ready for analysis.

Slide 8 - Time Feature Engineering

Once the data was cleaned, I performed feature engineering on the date field. From the cleaned date, I created four additional fields: year, month, month name, and day of the week. Why did I do this? Because a single date field is useful, but these additional features make temporal analysis much easier. Instead of repeatedly processing dates during analysis, I can directly group observations by year, month, or weekday. So this step converts one raw date field into reusable analytical dimensions.

Natural interview-style questions

Question: What is feature engineering?

Answer: Feature engineering means creating useful analytical variables from existing data. Here, I transformed the date into year, month, month name, and weekday features so that time-based analysis could be performed more easily.

Slide 9 - Final Manipulation Summary

At this stage, I summarized the final result of the data-preparation pipeline. I checked the final shape of the dataset along with the main quality indicators. This acted as the final audit point before the dataset was passed to the dashboard and exploratory analysis. So, at the end of the manipulation stage, I had a structured and cleaner analytical dataset that could be used safely for further analysis.

Natural interview-style questions

Slide 10 - Professional Data Preparation Conclusion

To summarize the data-preparation part, I followed a reproducible pipeline. I inspected the data first, then cleaned it systematically, converted important fields, normalized text, removed duplicate and empty records, created time-based features, and finally validated the output. The main objective was not to change the meaning of the observations. The objective was to make the data more consistent, comparable, and suitable for analysis. This cleaned dataset then became the foundation for the Bird Monitoring Dashboard.

Natural interview-style questions

Slide 11 - Dashboard Introduction

Now let us move from data preparation to the dashboard analysis. The dashboard contains 33,251 observations across 78 unique species. The purpose of the dashboard is not simply to display charts. Instead, I designed the analysis around a sequence of questions: Which species are observed most frequently? How are these birds being identified? How long are the observation intervals? How is sex represented in the observations? And finally, how do species contribution and identification methods connect with each other?

Natural interview-style questions

Speaking tip: pause after the KPI figures so the audience can absorb the scale of the dataset.

Slide 12 - Top 10 Species + Identification Method

The first pair of charts focuses on species frequency and identification method. First, I counted observations for each scientific name and selected the top ten species. The most frequently recorded species is *Cardinalis cardinalis*, with 8,514 observations. It is followed by *Passerina cyanea* with 3,860 observations and *Spizella pusilla* with 3,542 observations. On the right side, we can see the bird identification methods. Singing is clearly the dominant method, with 23,367 records, which represents about 70.3 percent of all observations. Calling contributes 6,218 observations and Visualization contributes 3,653. So the important insight from these two charts is that the observation volume is concentrated around a relatively small group of species, and a large proportion of observations are identified through singing.

Natural interview-style questions

Question: What is the analytical value of these charts?

Answer: They tell us which species are driving the observation dataset and also which identification methods are contributing most to those observations.

Slide 13 - Observation Interval + Overall Sex Distribution

The next pair of charts looks at observation structure. The first chart shows observation interval length. The most common category is the 0 to 2.5 minute interval, with 16,920 observations, or about 50.9 percent of the dataset. So we can see that a large portion of the observations come from relatively short intervals. The second chart shows overall sex distribution. Here, the largest category is Undetermined, with 20,269 records, representing about 61 percent. Recorded Male observations are 12,440, while Female observations are 542. Therefore, we should be careful when interpreting sex-related results. These figures represent recorded observations in the dataset, not the actual biological population.

Natural interview-style questions

Question: Can we say that there are more male birds than female birds?

Answer: No. We should not make that conclusion from this dashboard. The dashboard shows recorded observations, and a large proportion of sex records are undetermined. Differences in observation effort and detection can also influence these numbers. So this is an observation pattern, not a population estimate.

Speaking tip: emphasize that recorded observations are not the same as biological population estimates.

Slide 14 - Species-wise Sex Distribution + Sex Ratio

Now I am going one level deeper into the sex analysis. First, I filtered the dataset to Male and Female observations and grouped them by scientific name. Then I selected the top ten species based on total recorded Male and Female observations. We can see that *Passerina cyanea* has 3,700 recorded Male observations compared with only 80 Female observations. For *Cardinalis cardinalis*, we see 825 Male observations and 297 Female observations, which is a relatively larger Female presence compared with some of the other species. I also calculated a Male-to-Female ratio. For example, *Passerina cyanea* has a ratio of 46.25. In cases where there are zero Female observations, I did not create an infinite ratio. Instead, the dashboard reports the ratio as N/A. So the bar chart shows the absolute difference, while the ratio provides a standardized numerical comparison.

Natural interview-style questions

Question: Why did you use both a chart and a ratio?

Answer: Because they answer two slightly different questions. The chart shows the actual observation volume, while the ratio helps compare the imbalance numerically across species.

Slide 15 - Species Contribution + Identification Method by Species

The final pair of charts connects species contribution with identification method. First, I ranked all species based on observation count. I kept the top five species as separate categories and grouped the remaining species into Other Species. The top five species together contribute 20,538 observations, which is about 61.8 percent of the complete dataset. Among them, *Cardinalis cardinalis* alone contributes 8,514 observations, or about 25.6 percent of all observations. The second chart then breaks down those leading species according to identification method. Singing is dominant for the leading species, but the exact mix differs from species to species. For example, *Cardinalis cardinalis* has substantial Calling and Visualization observations in addition to Singing, while *Passerina cyanea* and *Spizella pusilla* are more concentrated in Singing-based records.

Natural interview-style questions

Question: Why did you group the remaining species into Other Species?

Answer: Because displaying every species separately would make the visualization difficult to read. By keeping the top five separate and grouping the rest, I can focus clearly on the species contributing the largest observation volume.

Speaking tip: explain why combining magnitude and ratio gives a stronger comparison.

Slide 16 - Overall Dashboard Conclusion

When I look at the dashboard as a complete analytical story, it starts with overall observation volume and species dominance. Then it introduces identification method. After that, it looks at observation intervals and overall sex composition. Then it goes into species-level sex comparisons and finally connects species contribution with identification method. So each chart answers a specific question, and the charts are connected rather than being isolated visualizations.

Natural interview-style questions

Slide 17 - Final Conclusion / Closing

To conclude, this project contains 33,251 bird observations covering 78 scientific species. The analysis shows that observation volume is concentrated in several species, Singing is the dominant identification method, short observation intervals are the most common, and Undetermined is the largest sex category. At the species level, recorded Male observations are often much higher than recorded Female observations. However, these are patterns in recorded observations and should not be interpreted as direct estimates of bird abundance or biological sex ratios. The main strength of this project is that I did not stop at creating individual charts. I created a complete analytical workflow: data preparation, cleaning, validation, feature engineering, dashboard analysis, and interpretation. The final dashboard therefore converts a large observation dataset into a set of focused questions and useful insights. Thank you.

Natural interview-style questions

Speaking tip: finish by connecting the visuals back to the original analytical questions.

Quick Practice Formula

For any chart, use this four-step pattern so your explanation sounds analytical and natural:

1. What was the question? - State what you wanted to understand.
2. What did I do? - Briefly explain the grouping, filtering, or calculation.
3. What did I find? - Give the most important insight and, when useful, one supporting number.
4. Why does it matter? - Explain what the result means for the project.

*Example: "The question here was which species were observed most frequently. To answer that, I grouped the data by scientific name and selected the top ten species. I found that *Cardinalis cardinalis* had the highest number of observations, with 8,514 records. This matters because it shows that a relatively small number of species contribute a large portion of the observation volume."*

Final reminders while recording

Do not read every word on the slide. Use the chart title as your cue and tell the story in your own voice.

Say the number only when it supports the insight. Avoid listing every bar or category.

Use phrases such as "the key insight is", "what this tells us is", and "we should be careful not to over-interpret this".

When discussing sex results, clearly say that the dashboard reflects recorded observations, not direct population estimates.